HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
Office for International Study Programs

CO3091 Logic Design Project

Project Progress Report

Step Detection System

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1 Introduction

Dear teacher,

In this paper, we present a step detection and measurement algorithm that integrates data from two key sensors: a foot pressure sensor (FSR) and a three-axis accelerometer (Oxyz). The main objective of this approach is to accurately identify each walking step by combining information about ground contact force and motion dynamics. While many traditional pedometers rely solely on accelerometer data, this method improves precision by verifying that a step only occurs when both movement and pressure thresholds are satisfied simultaneously.

The algorithm begins by preprocessing the raw signals to remove noise, then analyzes the acceleration along the X, Y, and Z axes to detect rhythmic motion patterns. Simultaneously, it monitors foot pressure to determine the exact moment the foot strikes or lifts from the ground. By correlating these two data sources, the algorithm minimizes false detections caused by minor vibrations, sudden hand movements, or surface irregularities.

Such an integrated system is particularly useful in health monitoring devices, rehabilitation training, and robotic locomotion systems, where accurate step detection is crucial. Furthermore, the algorithm can be extended with adaptive thresholds or machine learning models to personalize detection for each user's walking style.

Thank you for taking the time to read this introduction. We hope this work inspires your exploration of sensor fusion techniques and contributes to more reliable movement tracking in your future projects.

Group 8,

Student Name	**Student ID**	**Task**	**Progress**
Vũ Tiến Vinh	2353334	Schematics, PCB, Hardware	-%
Nguyễn Viết Tấn Vương	2353345	Support, Hardware	-%
Phạm Quốc Vương	2353346	Arduino, Algorithms, Software	-%

2 Theory

2.1 Human Walking Dynamics

Walking is a repetitive biomechanical process involving a sequence of coordinated limb and body movements. Each gait cycle is divided into two main phases: stance (when the foot is in contact with the ground) and swing (when the foot moves forward in the air). During walking, the body's center of mass oscillates both vertically and horizontally, resulting in cyclical patterns of acceleration and ground reaction force. These oscillations reflect the rhythmic nature of human motion and can be used as measurable indicators of step occurrence.

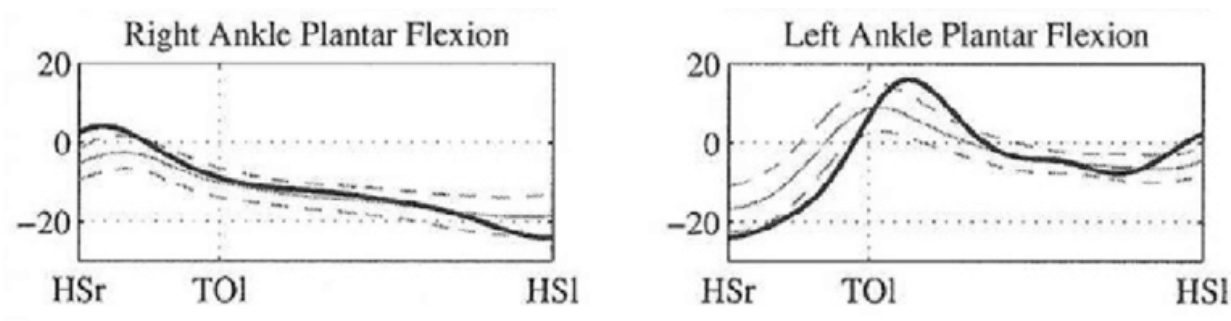


Figure 1: Measured and simulated motions of lower limb

The periodicity of gait makes it suitable for quantitative analysis: each cycle typically lasts 0.4–0.8 seconds, depending on walking speed. By measuring how acceleration and force vary within this cycle, it becomes possible to detect and count individual steps accurately.

2.2 Motion Analysis via Acceleration

Acceleration is a key physical parameter that describes the rate of change in velocity over time. In three-dimensional motion, it can be represented by components along the x, y, and z axes:

$$A_{\text{total}} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

This total acceleration magnitude provides a scalar representation of body motion. When a person is standing still, A_{total} remains close to the gravitational constant ($g = 9.81$ m/s^2 or $1g$). During walking, it oscillates periodically above and below this baseline.

Each peak in acceleration corresponds to a change in body momentum—usually occurring when a foot strikes or pushes off the ground. By analyzing these periodic fluctuations, one can infer walking frequency and identify the timing of steps. However, external vibrations or upper-body movements can cause noise, so acceleration alone cannot always distinguish genuine steps from false signals.

2.3 Ground Reaction Force and Contact Phases

The ground reaction force (GRF) represents the opposing force exerted by the ground whenever a foot is in contact with it. This force varies predictably within each step: it rises sharply at heel

strike, stabilizes as weight is supported, and drops as the foot lifts during toe-off.

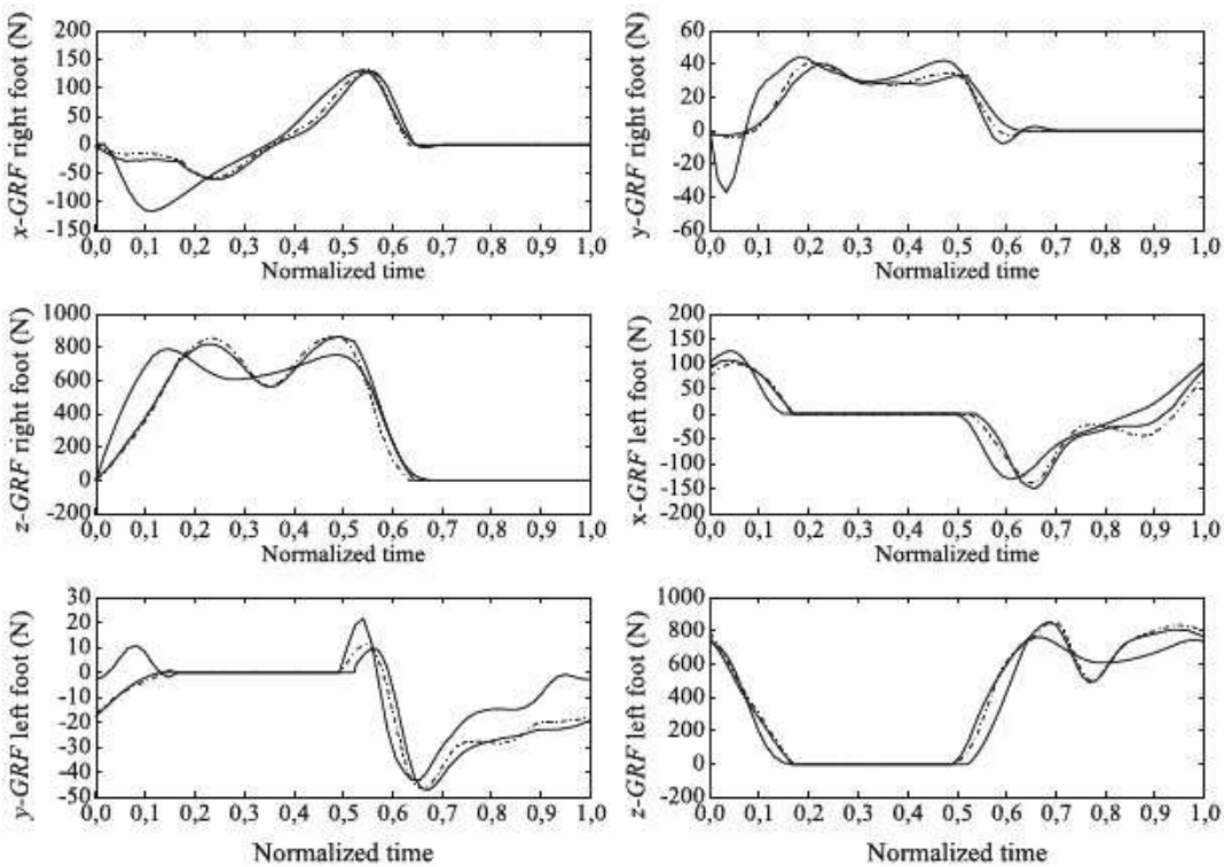


Figure 2: Ground reaction forces of both feet during spatial motion

The GRF curve typically displays two characteristic peaks—one from the heel impact and another from the push-off. The interval between these peaks corresponds to the stance phase of walking.

By setting a force threshold F_{th} one can distinguish between periods when the foot is on the ground ($F > F_{\text{th}}$) and when it is lifted ($F \leq F_{\text{th}}$).

The combination of these two signals—force for contact confirmation and acceleration for dynamic motion—forms the foundation of a robust step detection model.

2.4 Signal Processing and Step Detection Principle

Real-world motion signals are often contaminated by high-frequency noise and random fluctuations. To enhance signal quality, a low-pass filter is applied to remove irrelevant components while retaining the primary walking frequency range (1–3 Hz). The filtered data are then analyzed to identify key features such as amplitude change and threshold crossings.

A step is recognized when both conditions below are met within a defined time interval:

$$(\Delta A > A_{\text{th}}) \wedge (F > F_{\text{th}})$$

where A_{th} is the motion threshold and F_{th} is the contact threshold.

This dual-condition approach ensures that only genuine walking steps—characterized by concurrent movement and contact—are recorded, minimizing false positives from random vibrations or posture shifts.

3 Methodology

To address the limitations of single-sensor approaches and eliminate false positives (e.g., stomping in place), this system employs a multi-sensor fusion strategy. The methodology combines inertial data from an MPU-6050 with kinetic data from a Force Sensitive Resistor (FSR) using a Finite State Machine (FSM).

3.1 Inertial-Based Swing Phase Detection

Instead of calculating the full trajectory via double integration (which is prone to drift), the system utilizes the accelerometer to detect the dynamic **Swing Phase** of the gait cycle. The 3-axis acceleration data (a_x, a_y, a_z) is fused into a single orientation-independent signal known as the Vector Magnitude (VM):

$$VM = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

A threshold-based approach is applied to this signal. The system identifies a "Swing" event when the VM exceeds a predefined threshold ($\tau_{swing} \approx 13.0 m/s^2$). This high acceleration characterizes the foot lifting off the ground and moving through the air, differentiating actual walking from static leg movements.

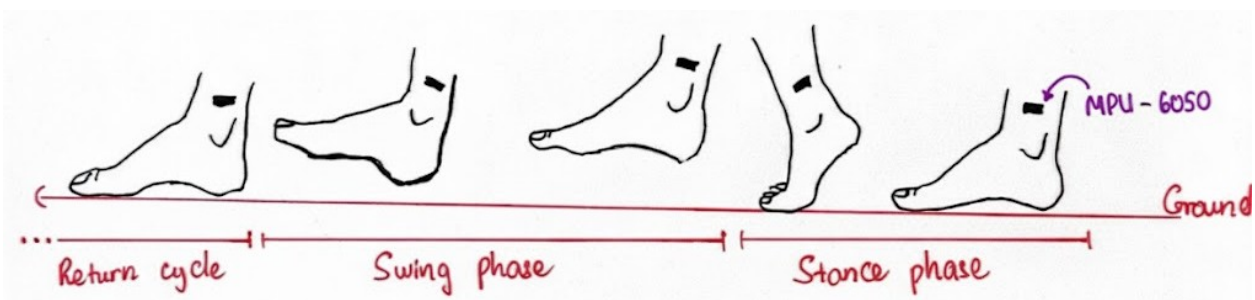


Figure 3: Status of step

3.2 Force-Based Gait Phase Transition

The FSR sensor provides ground truth for ground contact. Rather than using simple peak detection, the algorithm analyzes the rate of change (derivative) of the force signal to classify specific gait phases. The change in force (ΔFSR) is calculated between consecutive sampling intervals:

$$\Delta FSR_t = FSR_t - FSR_{t-1} \tag{2}$$

Based on the raw force (FSR_t) and its derivative (ΔFSR_t), the system classifies the gait into four distinct phases:

- **Heel-Off:** Foot lifting, characterized by a rapid drop in force ($\Delta FSR < -\delta_{limit}$).

- **Swing Air:** Foot in the air ($FSR \approx 0$ and High Acceleration).
- **Heel-Strike:** Heel impact, characterized by a rapid rise in force ($\Delta FSR > \delta_{limit}$).
- **Stance Flat:** Full foot contact with stable force ($\Delta FSR \approx 0$).

3.3 Sensor Fusion and State Machine Logic

The core innovation of this methodology is the "Swing-Prior-to-Impact" logic implemented via a Finite State Machine. This logic prevents false step counting caused by merely tapping the foot while sitting (stomping).

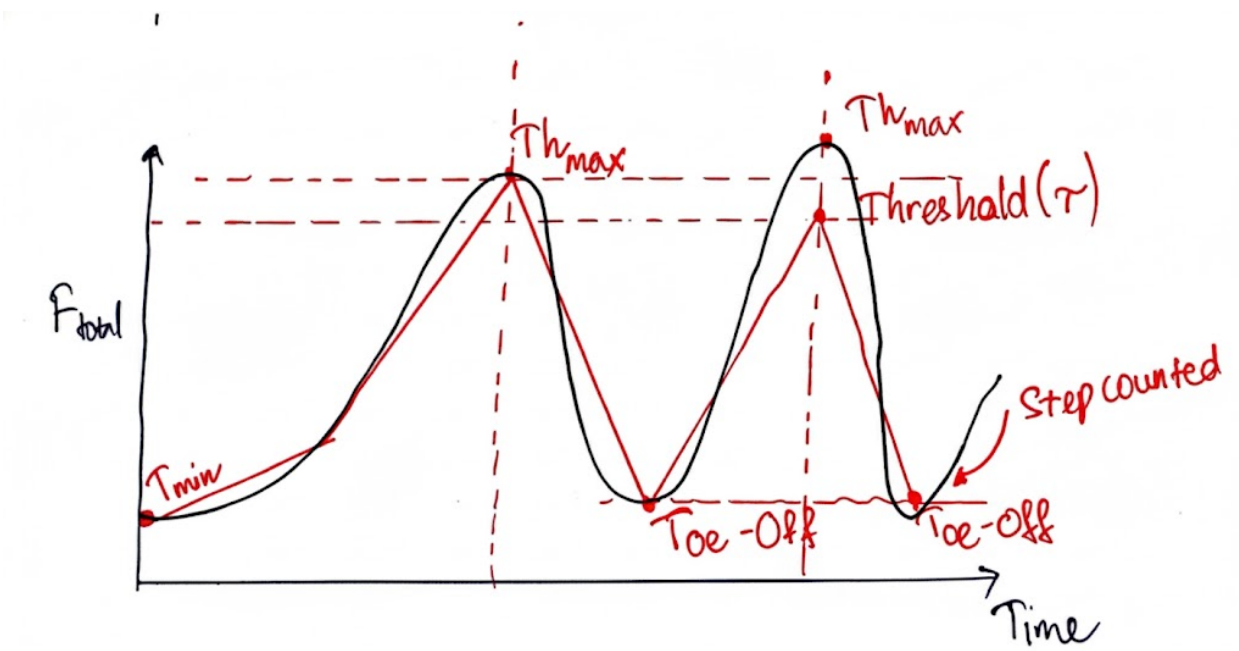


Figure 4: Graph of FSR simulation

The step counting algorithm follows a strict sequential requirement:

1. **Swing Priming:** The system continuously monitors the MPU-6050. If a valid Swing Phase is detected ($VM > \tau_{swing}$), a flag `swingPrimed` is set to TRUE.
2. **Impact Validation:** The system waits for a Heel-Strike event from the FSR.
3. **Step Registration:** A step is only incremented if:

$$Step = \begin{cases} Step + 1 & \text{if } (State = \text{Heel-Strike}) \wedge (\text{swingPrimed} == TRUE) \\ Step & \text{otherwise} \end{cases} \tag{3}$$

This fusion ensures that a physical step is recorded only when the biomechanical sequence of *Lift* $\rightarrow$ *Swing* $\rightarrow$ *Impact* is fully satisfied.

4 Design Spetification

4.1 BOM and Block diagram

The objective of this design is to create an ankle-worn step tracking device capable of accurately detecting walking motion while ensuring a compact structure and cost-effective component selection. Figure 1 lists the essential components used in the implementation.

Component (Model)	**Quantity**	**Key Specifications**
Transceiver circuit Bluetooth (HC-06)	1	Bluetooth 2.0+EDR; UART; 3.6–6V; 9600bps; ∼10m range
Microcontroller (Arduino Nano)	1	5V; 16MHz; 14 DIO (6 PWM); 8 ADC
Accelerometer Sensor (MPU6050)	1	3-axis accel + 3-axis gyro; I2C; 3.3–5V
Force Sensing Resistor (DF9-40)	1	Flexible FSR; range up to 5kg; analog output
Boost Converter (MT3608)	1	Input 2–24V; Output up to 28V; Max 2A; >93% efficiency
Micro SD Card	1	4GB
Micro SD Module	1	SPI interface; 3.3V logic; compact board
Battery (AA)	2	3.7V
Charging module (TP4056)	1	5V – 1A
LED	3	5V; 220–330Ω
Push Button	1	6mm × 6mm × 5mm

Table 1: List of required components for the design

The principal objective of this design is to interface motion and pressure sensors with the microcontroller to perform step detection and data transmission. Consequently, the overall schematic structure of the system is presented in Figure 5.

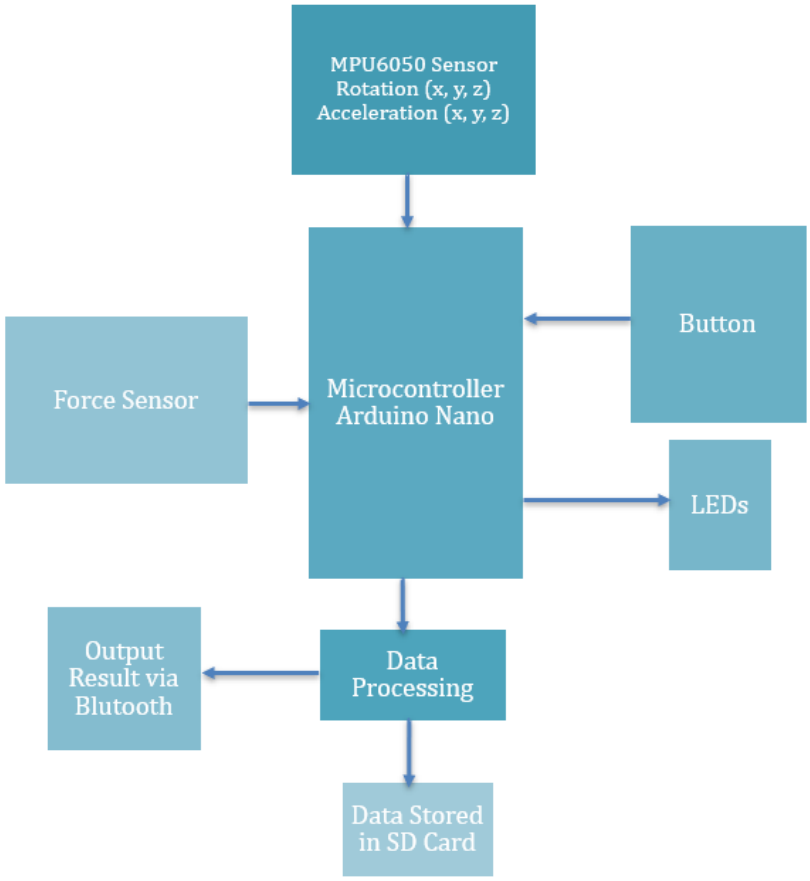


Figure 5: Schematic block diagram of the system

The Step Detection System operates by collecting and processing motion and pressure data to identify each step accurately. The MPU6050 sensor measures the angular rotation and linear acceleration along the X, Y, and Z axes, while the force sensor detects the applied pressure on the foot. Both signals are transmitted to the Arduino Nano microcontroller, which acts as the control center of the device.

Inside the microcontroller, data are filtered and analyzed through step-detection algorithms. The push button allows the user to control system states such as start, pause, and stop, and the LED indicators provide real-time visual feedback corresponding to each mode.

Processed data are stored in the microSD card for later review and simultaneously sent via the Bluetooth module to a mobile device for real-time monitoring. Through this workflow, the Step Detection System ensures precise measurement, reliable step counting, and convenient user interaction.

4.2 Design Layout

The system design section presents the overall hardware configuration of the Step Detection System. It includes both the schematic circuit diagram and the physical breadboard implementation, illustrating how each hardware component is interconnected and powered.

4.2.1 Schematic Circuit Diagram

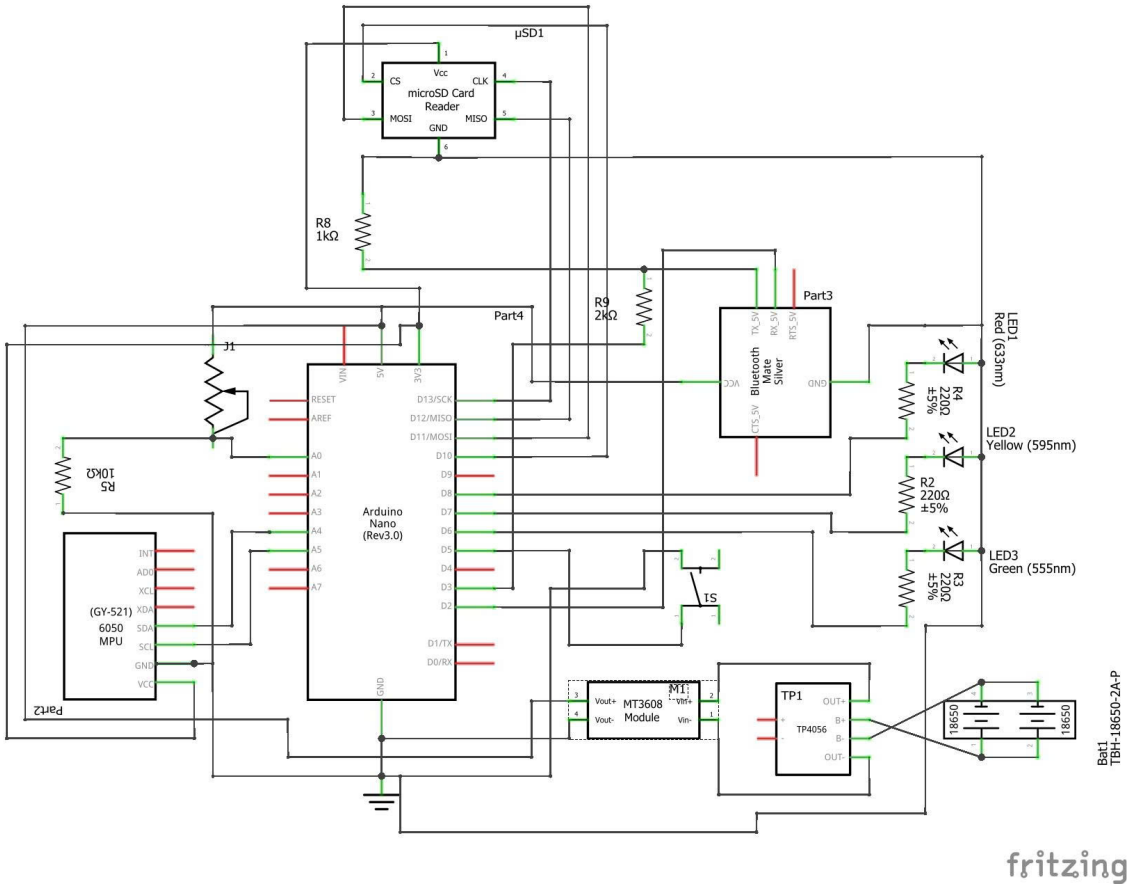


Figure 6: Schematic circuit diagram of the system

Figure 6 shows the complete schematic of the Step Detection System. The Arduino Nano serves as the central control unit, interfacing with all peripheral components. The MPU6050 accelerometer and gyroscope module connects to the Nano through the I^2C communication lines (SCL and SDA), allowing continuous motion sensing. The force sensing resistor (FSR) is connected to one of the analog input pins, providing a variable voltage signal proportional to the applied pressure on the foot.

The MT3608 boost converter regulates and increases the supply voltage to stable 5V for the entire circuit. The TP4056 charging module manages battery charging and protection, ensuring a reliable power supply. The HC-06 Bluetooth module is linked via UART pins (TX and RX) for wireless data transmission to a mobile device, while the microSD module connects through SPI pins for step data logging. Additionally, three LEDs are connected to digital output pins through

current-limiting resistors to indicate system states, and a push button is connected to a digital input for user mode control.

4.2.2 Breadboard Implementation

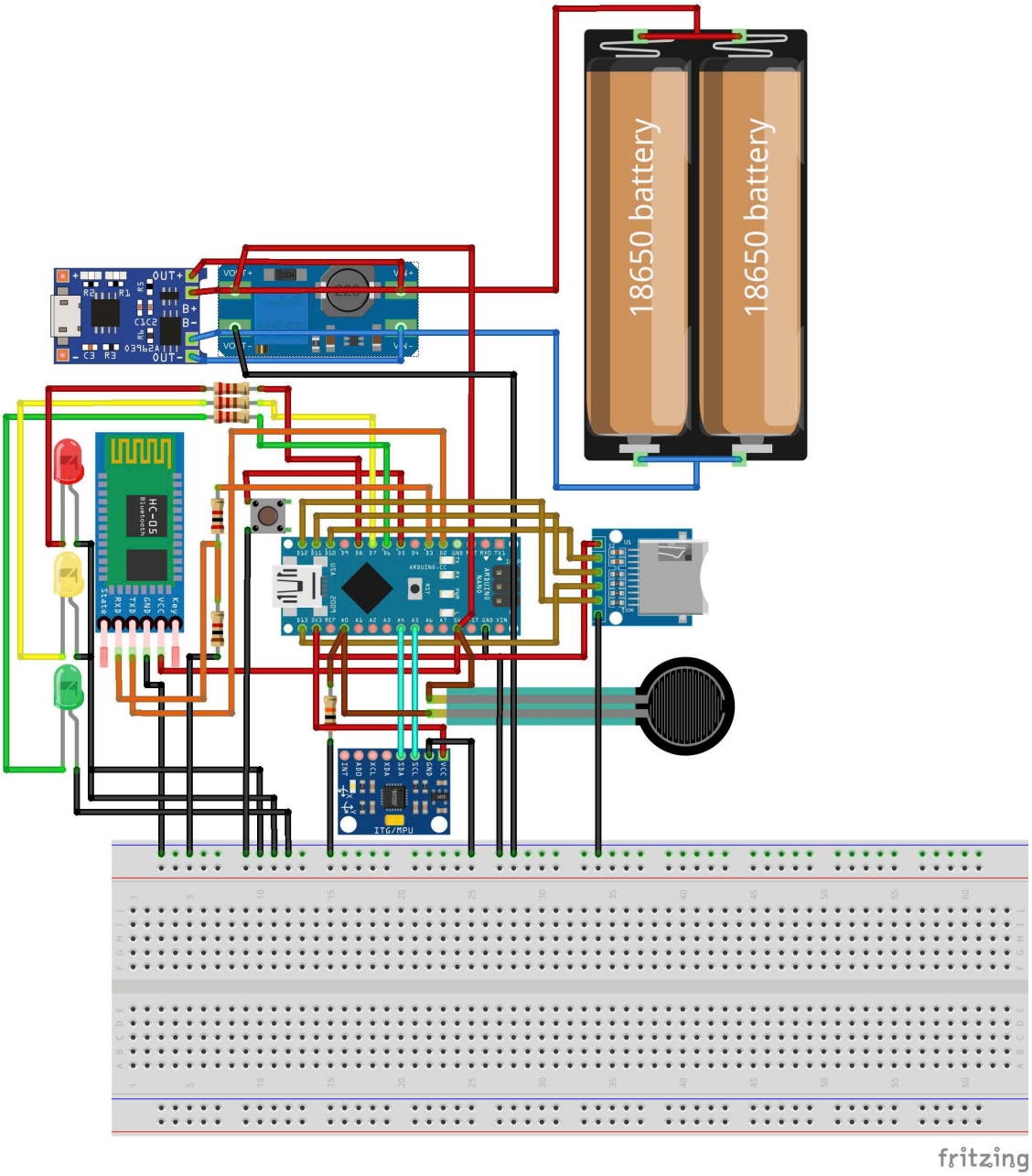


Figure 7: Breadboard layout of the system

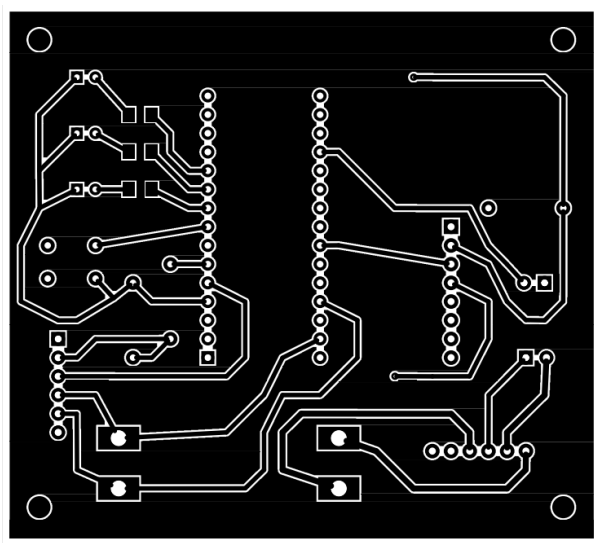
Figure 7 presents the breadboard layout used to prototype and test the system. The Arduino Nano is positioned at the center, acting as the hub that connects all other components. The MPU6050 sensor module is connected through the 3.3V and GND rails, with SCL and SDA linked

to pins A5 and A4 of the Nano, respectively. The force sensor's output line is connected to analog input A0, enabling real-time pressure measurement.

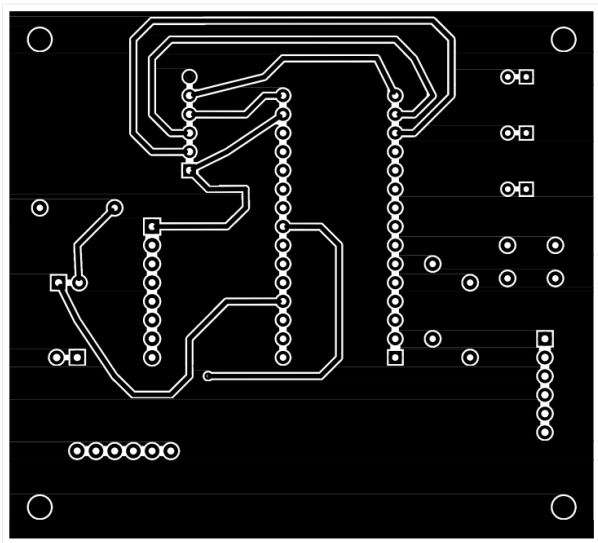
Three LEDs are placed on the right section of the board and connected to digital pins D9, D10, and D11 via 220 ohm resistors to display system states. A push button is wired to pin D2 with a pull-down resistor, allowing manual control of start, pause, or stop operations. The Bluetooth HC-06 module is connected to pins D2 and D3 for data transmission, and the microSD card module is wired via the SPI interface (pins D10–D13). The power supply is delivered through the TP4056 charging circuit and MT3608 boost converter, ensuring stable voltage to both logic and peripheral modules.

This physical setup replicates the schematic connections and provides a reliable test environment for verifying data acquisition, control response, and communication functionality.

4.2.3 PCB Layout



(a) Copper Top Mirror Layer



(b) Copper Bottom Layer

Figure 8: Top and Bottom Copper Layers

4.3 Testing Product

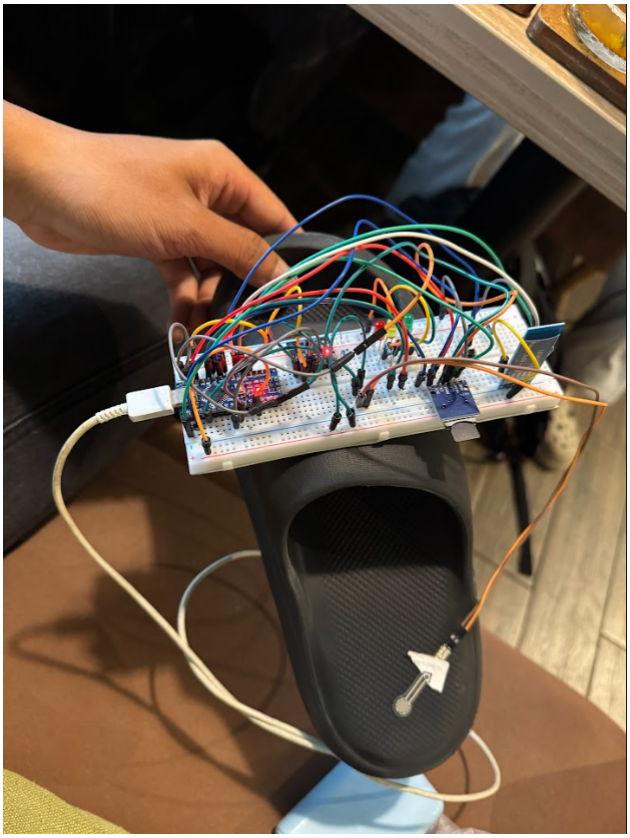


Figure 9: Simulation

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